Title	System block	
Size: E	Document Number: <Doc>	Rev: <RevCode>
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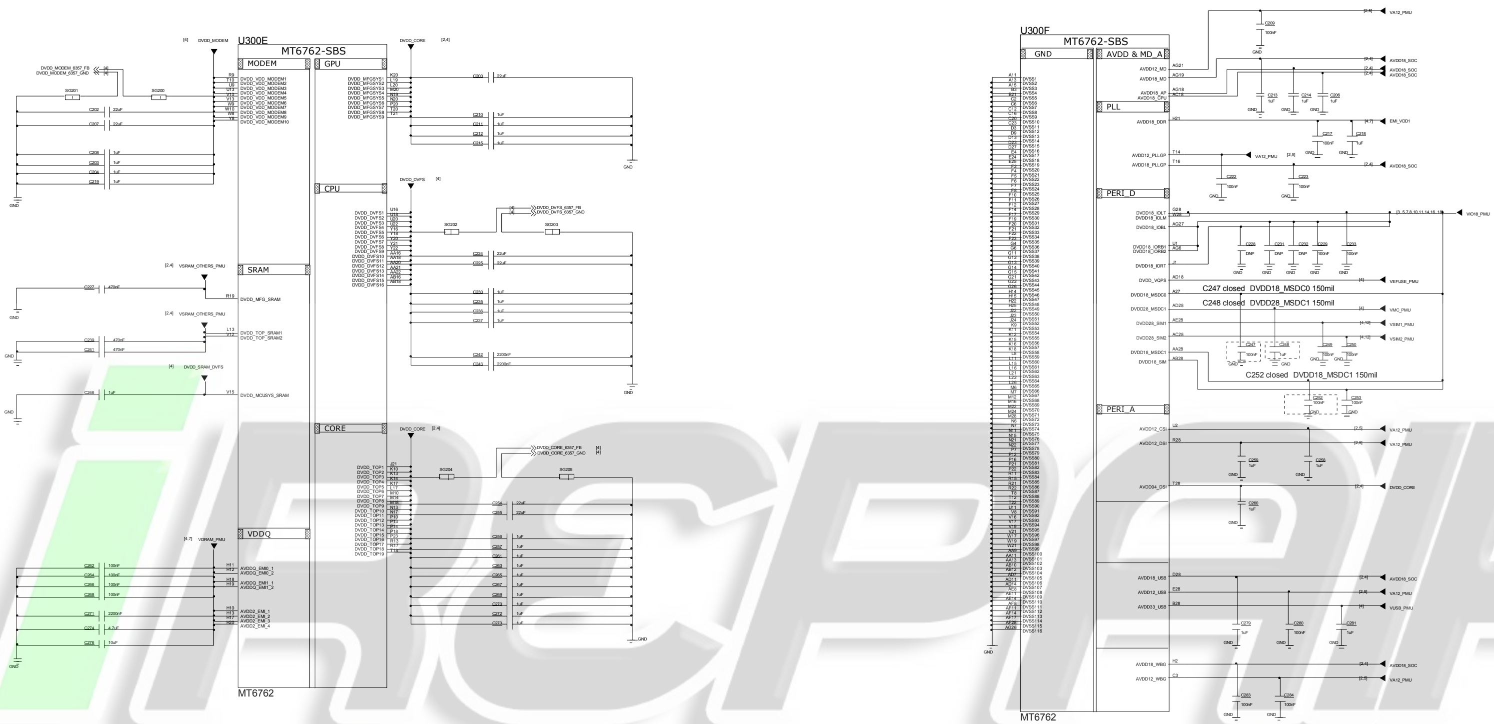

Title	MT6761:POWER	
Size: E	Document Number: M6102-FIJI	Rev: V10
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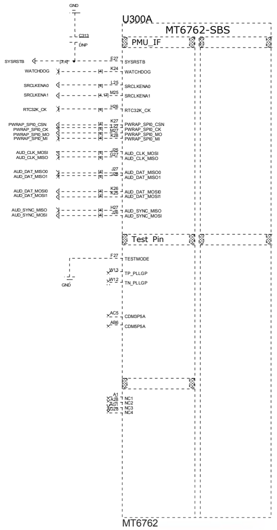
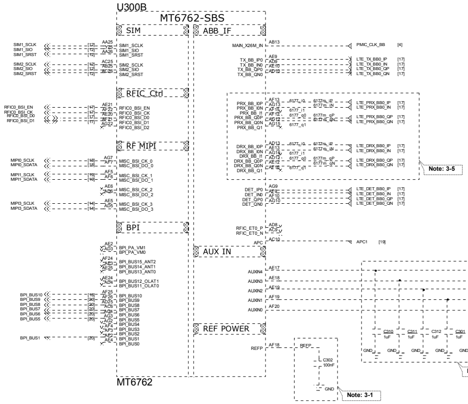
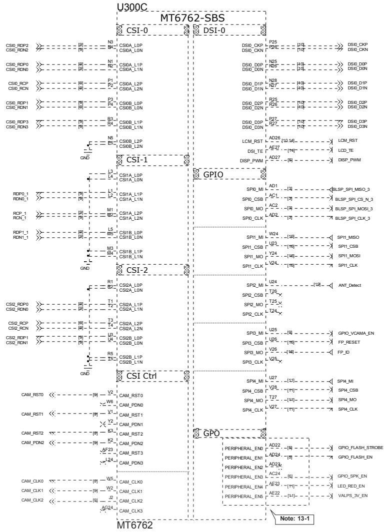
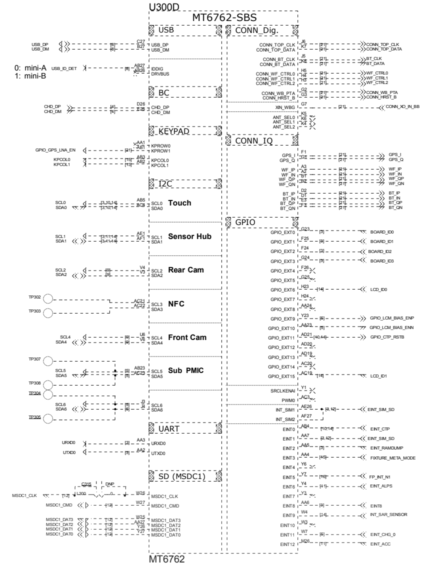








Schematic design notice of "12_BB_1" page.

Note 3-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 3-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

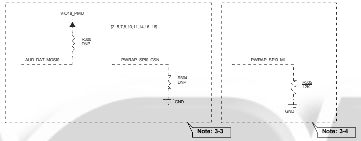
Note 3-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

	PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
	default=PU	default=PD	AP_JTAG	MD_JTAG
USE CASE	HI	LO	N/A	N/A
	HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3
	LO (by ext. PD)	LO	SPI0+EINT8	N/A
	LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

Note 3-4: PWRAP_SPI0_MI is DDR type feature in bootstrap

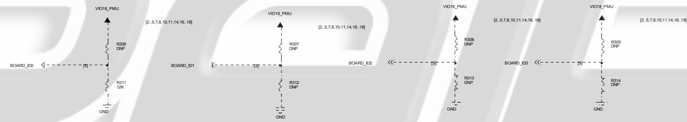
	PWRAP_SPI0_MI	Booting interface	
	default=PU	DDR	MSDC0 pin mux
USE CASE	LO (by ext. PD)	LPDDR3	follow LP3 Ref SCH.
	HI	LPDDR4X	follow LP4X Ref SCH.

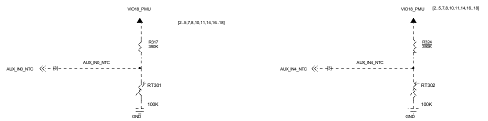
Note 3-5: Please set unused IQ pins in NC



Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND



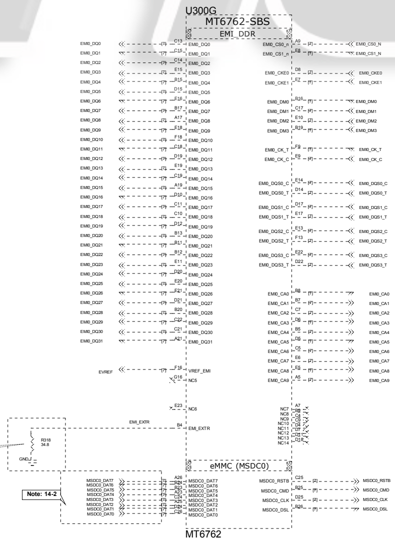


Thermistor to sense AP temperature

1. RT301 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

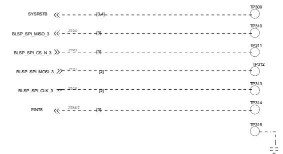
Thermistor to sense Board temperature

1. RT302 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

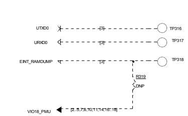


R318 please select 34.8 ohm (1%) resistor

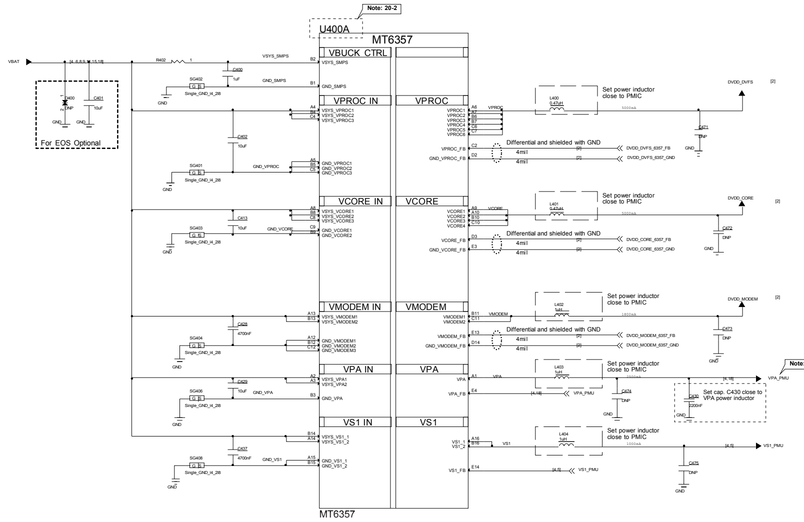
JTAG



UART



Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux



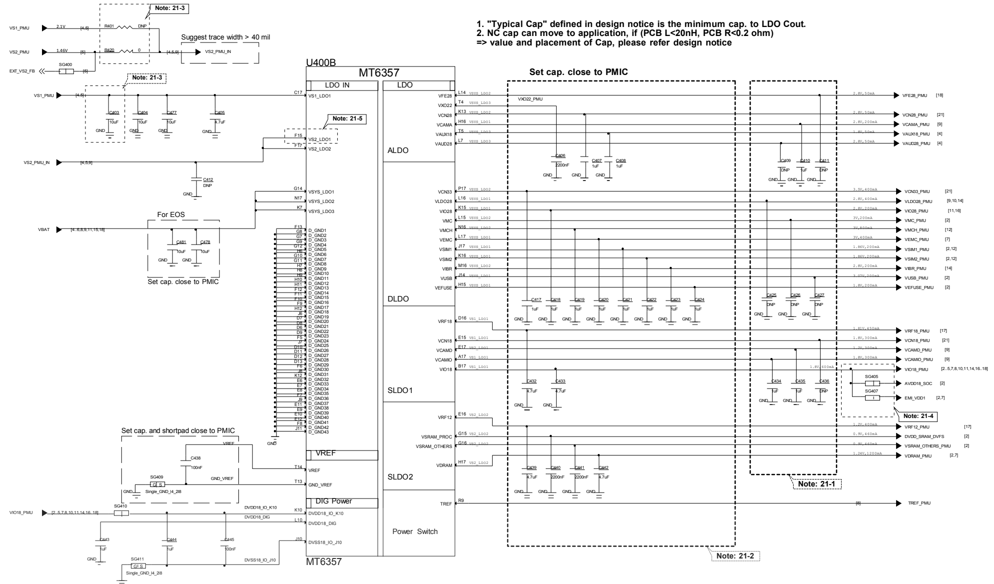
Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C2040, please choose 0402 size

Note 20-2: PMIC Part number notice for MT6765/62/61 platform

MTK Platform	PMIC
MT6765 / 62	MT6357CRV
MT6761	MT6357MRV

1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2 ohm)
=> value and placement of Cap, please refer design notice



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

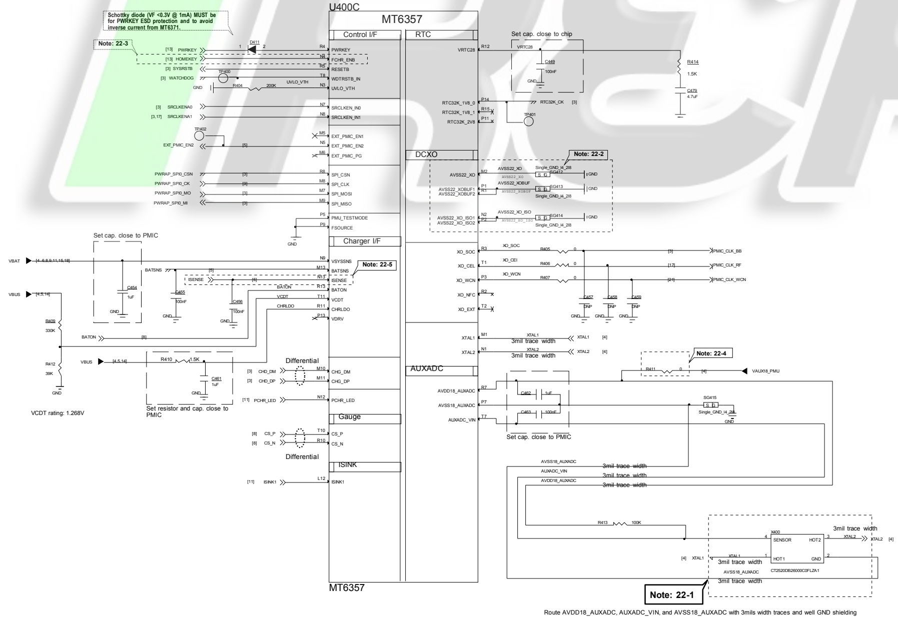
Note 21-3: Ext Buck BOM option

	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C403	10uF(0402)	22uF(0603)
R420	0-ohm , 0603	NC
R401	NC	0-ohm , 0603

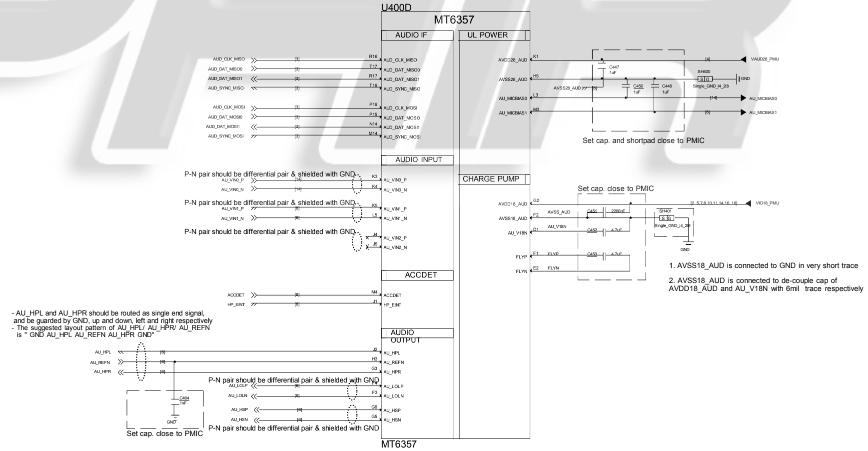
Note 21-4: Please set SG405 and SG407 close to C433, making star connection among VIO18_PMU, AVDD18_SOC, and EMI_VDD1 near to LDO cap. C433

Please also refer to MT6357 design notice for further detail design information

Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) >= 1.3 V



Route AVDD18_AUXADC, AUXADC_VIN, and AVSS18_AUXADC with 3mils width traces and well GND shielding



Schematic design notice of "22_POWER_MT6357-IF"

Note 22-1: Please implement 2520 & 2016 Size TMS PCB co-layout.
Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide

Note 22-2:
1. Please Connect P1 and R1 ball first and then to GND
2. Please Connect P2 and N2 ball first and then to GND
3. Please connect DCXO GND to main GND by independent L1-2 GND via.;
DO NOT connect it through L1 GND

Note 22-3: Let floating if disable HOMEKEY function

Note 22-4: Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R411 can use 0 ohm to replace BEAD.

Note 22-5: Please connect to battery connector

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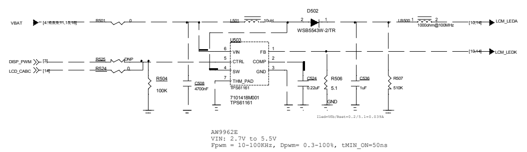
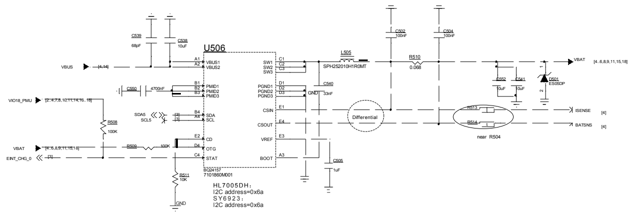
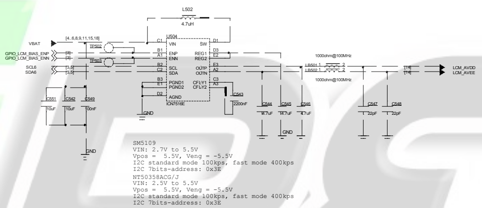
LCM Backlight LED driver



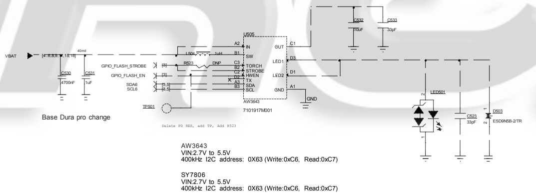
Charger



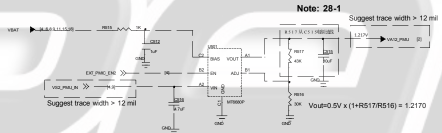
LCD Bias



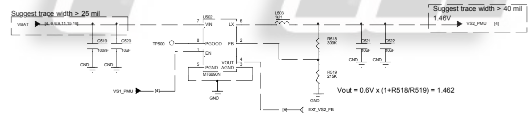
Flash LED 5V Boost



LDO for VA12

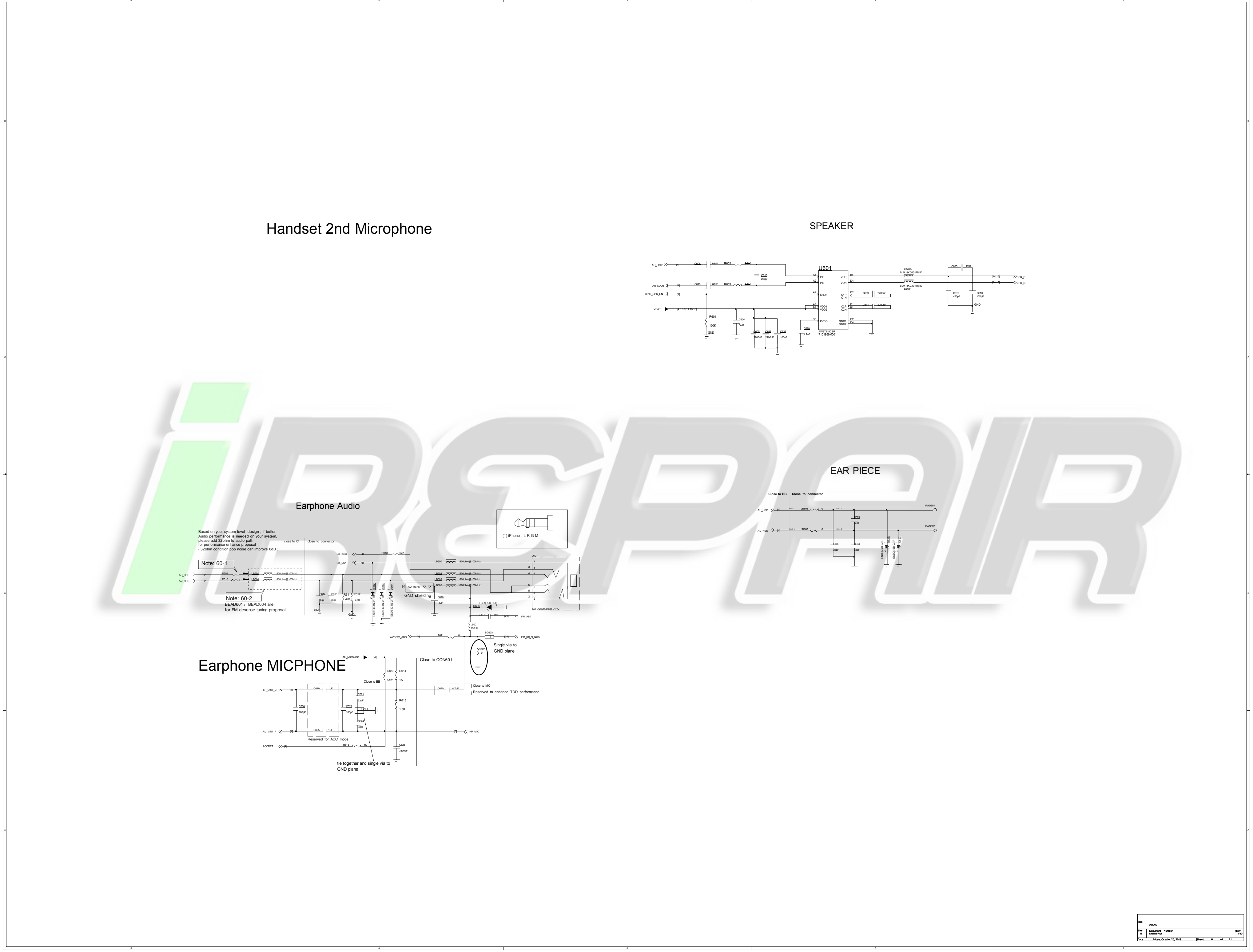


Ext. Bulk for VS2

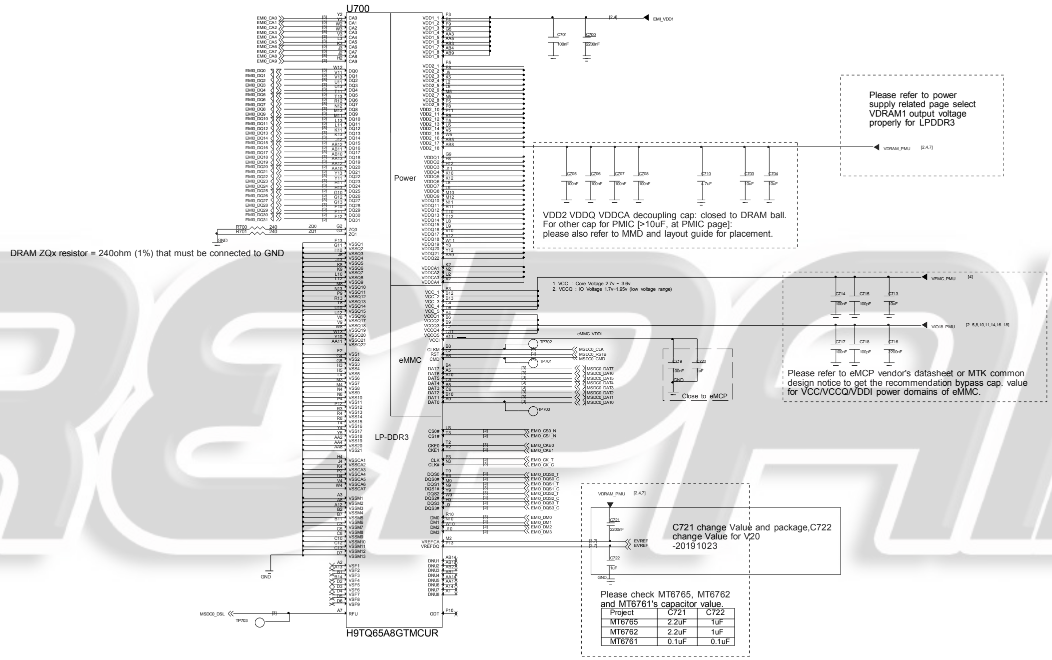


Schematic design notice of "28_POWER_ThirdParty-Power"	
Note 28-1:	VA12 Layout placement please close to AP
Note 28-2:	VS2 Buck Layout placement please close to PMIC MT6357
Note 28-3:	VCN33 LDO Layout placement please close to MT6631

Schematic design notice of "27_POWER_SubPMIC-HV powers" page:
Note 27-1: To minimize RF de-sense, it is recommended to reserve 0-ohm and 0402 cap for BOM fine tuning.
Note 27-2: To minimize RF de-sense, it is recommended to reserve 0-ohm and 0201 cap. for BOM fine tuning.
Note 27-3: C2705 could be replaced with C / 1 / uF / 50V + C / 1 / uF / 50V



Handset 2nd Microphone
SPEAKER
U601
EAR PIECE
Close to BB
Close to connector
Earphone Audio
Based on your system level design , if better
Audio performance is needed on your system,
please add 32ohm to audio path
for performance enhance proposal
(32ohm condition pop noise can improve 6dB)
Note: 60-1
Note: 60-2
BEAD601 / BEAD604 are
for FM-desense tuning proposal
(1) iPhone : L-R-G-M
GND shielding
Single via to
GND plane
Earphone MICPHONE
Close to CON601
Close to BB
Close to MIC
Reserved to enhance TDD performance
Reserved for ACC mode
tie together and single via to
GND plane



Project	C721	C722
MT6765	2.2uF	1uF
MT6762	2.2uF	1uF
MT6761	0.1uF	0.1uF

Title eMCP:LPDDR3&eMMC

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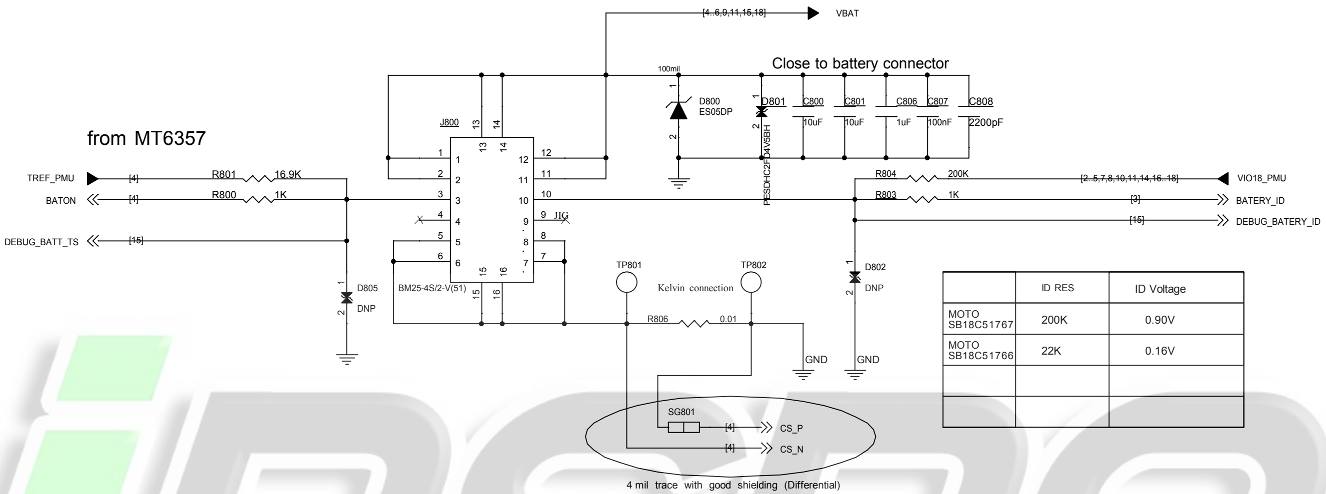
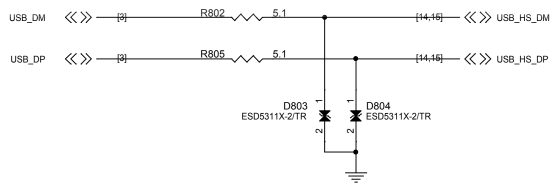
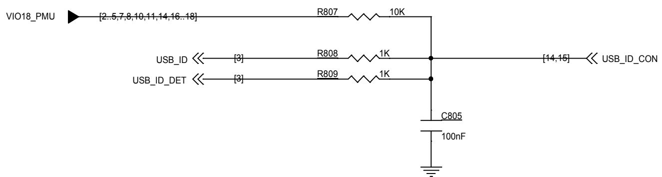
BATTERY CONNECTOR



	ID RES	ID Voltage
MOTO SB18C51767	200K	0.90V
MOTO SB18C51766	22K	0.16V





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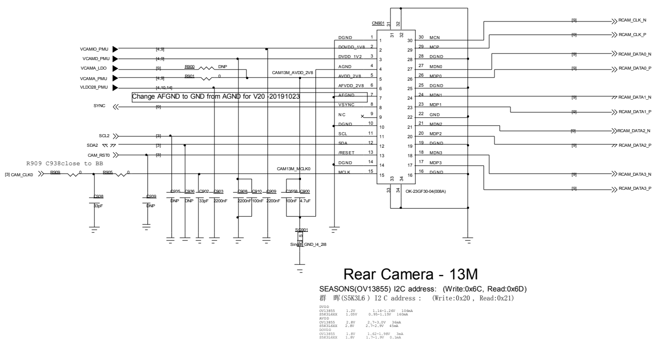
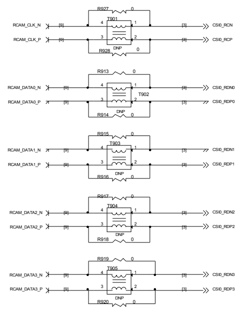
Page09:Rear-CAM-13M,2M
Front-CAM-5M

Rear-Main Camera(AF-13M)

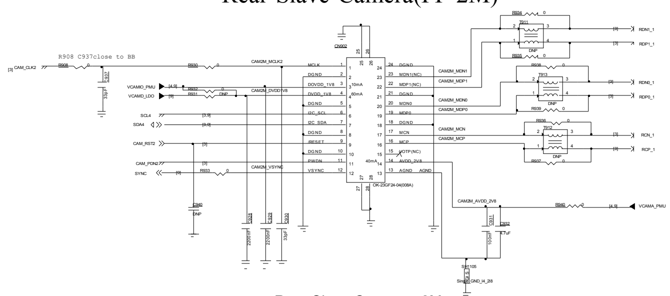




Rear Camera - 13M

SEASONS(OV13855) I2C address: (Write:0x6C, Read:0x6D)

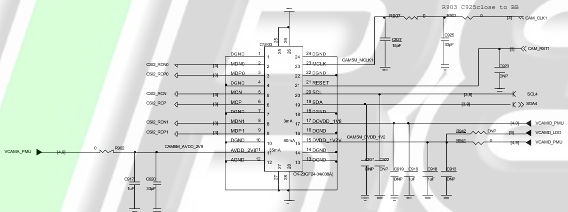
Rear-Slave Camera(FF-2M)



Rear Slave Camera - 2M

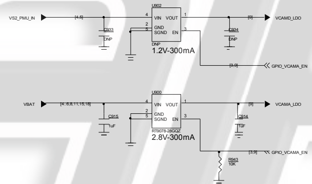
SEASONS(GC2375-C23Y0) I2C address: (Write:0x2E, Read:0x2F)

Front-FF-5M Camera Connector

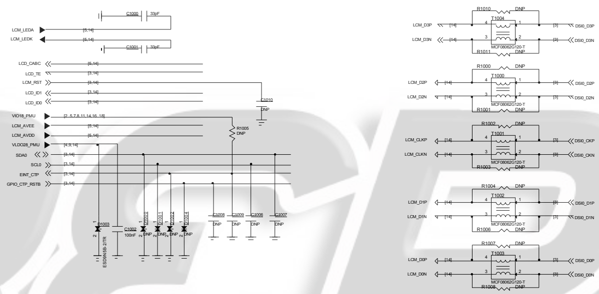


Front Camera - 5M

SEASONS(GC5035-WC1X0) I2C address: (Write:0x6E, Read:0x6F)



LCM Connector



GT1151 I2C address: 0X5D (Write:0xBA, Read:0xBB)
or 0x14 (Write:0x28, Read:0x29)

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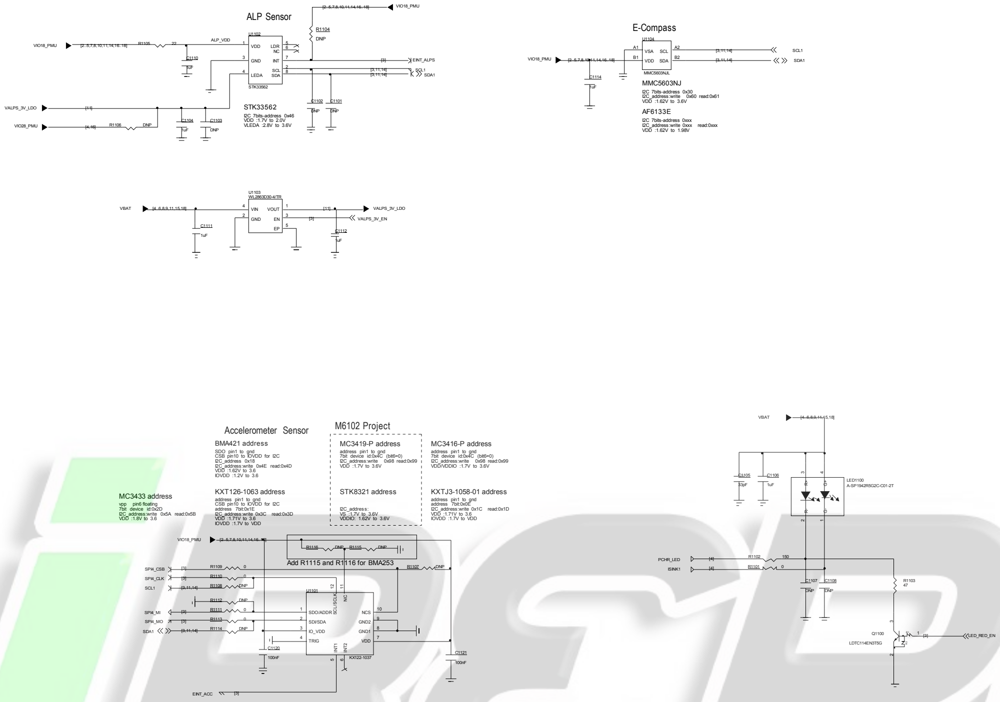

Title: Sensor		
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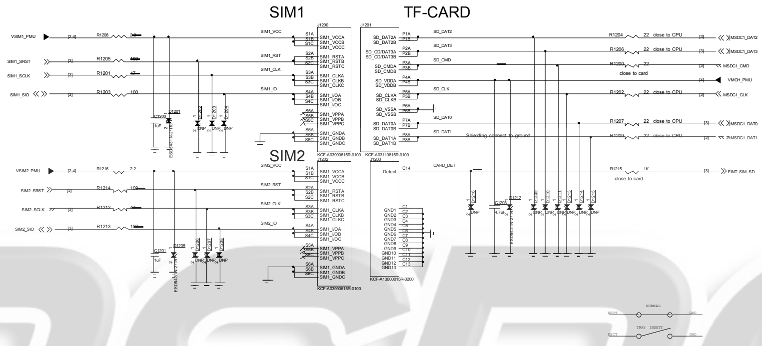
SIM and TF



Title: SIM&TF-Card		
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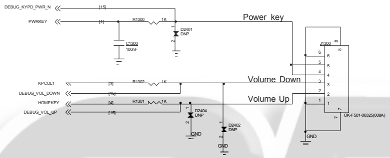
KEY

Volume Up : HOME Key / GND
Volume Down : KPCOL1/GND

DO NOT put pull-up resistor on PWRKEY



Title	Key-CON		
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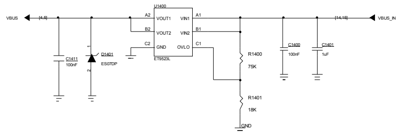
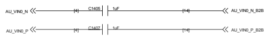
OVP

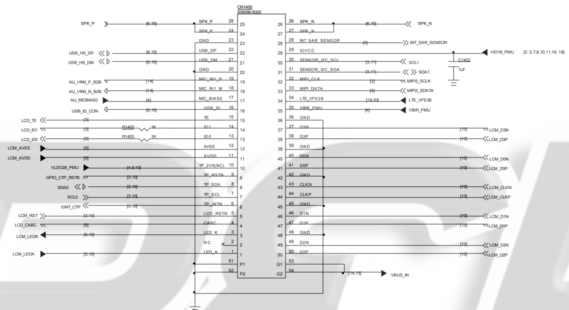


VIN_OVLO=VOVLO_TH x (1+R1/R2) = 1.2 * (1 + 75/18) = 6.2

Sub Connector

0xXX (Write:0xXX, Read:0xXX)





Title: OVP&CON		
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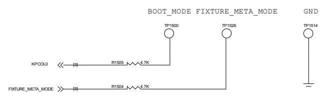
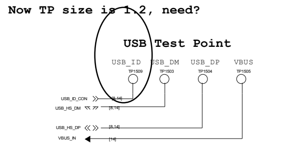
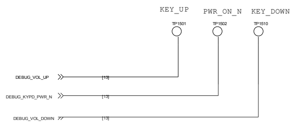
Download mode TP



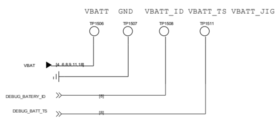
USB Test point

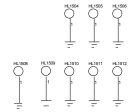


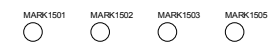
KEY Test point

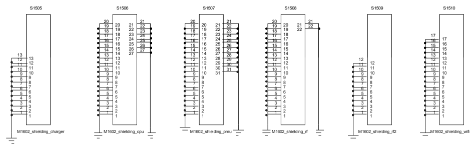


BATTERY Test point

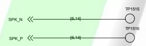




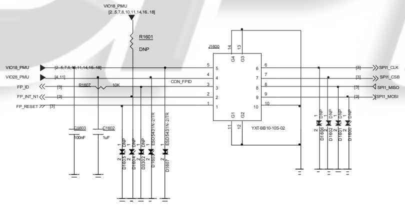




SPK Test point



Fingerprint



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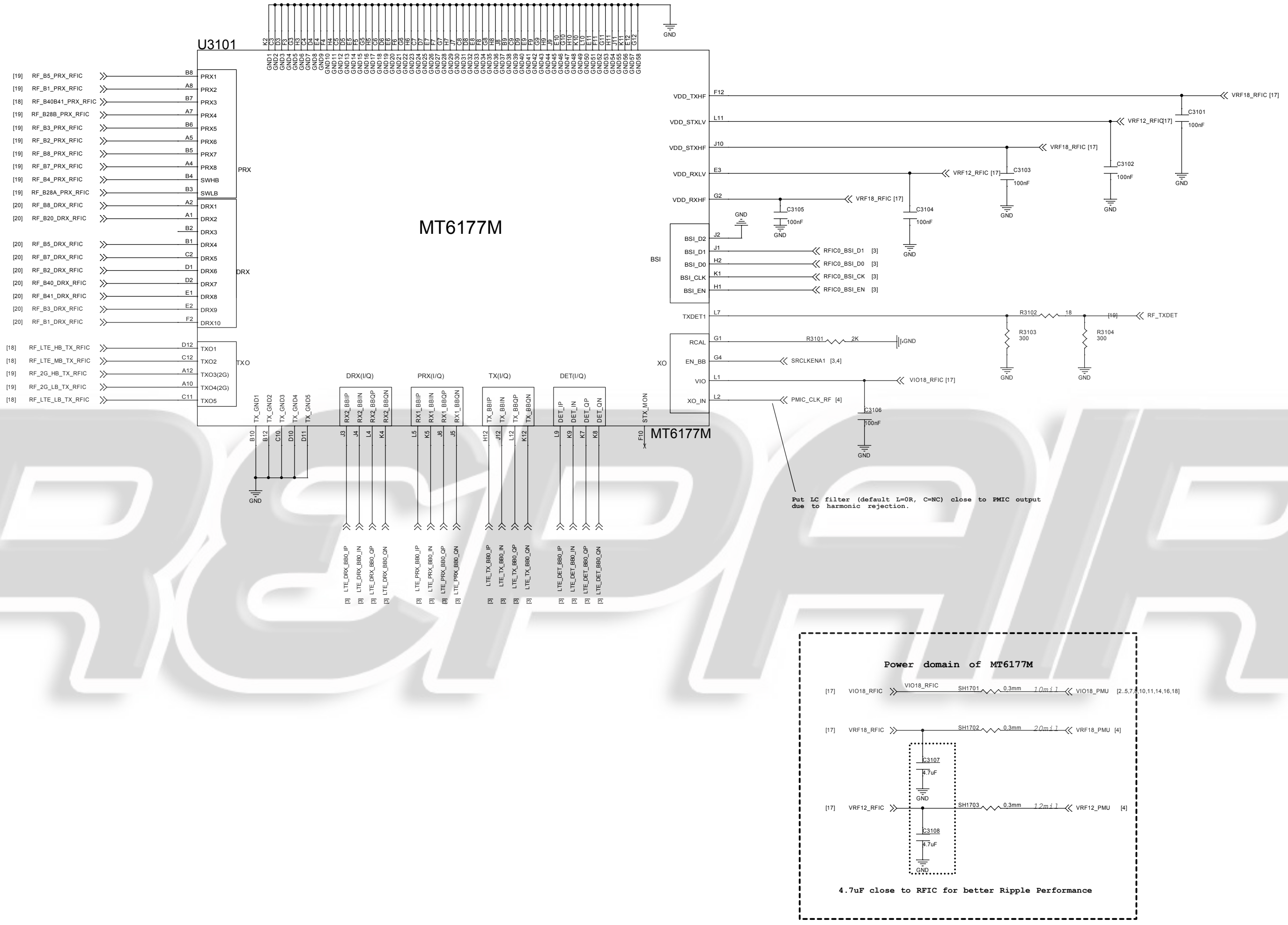

RF MT6177

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Size D	Document Number M6102-F1JI	Rev
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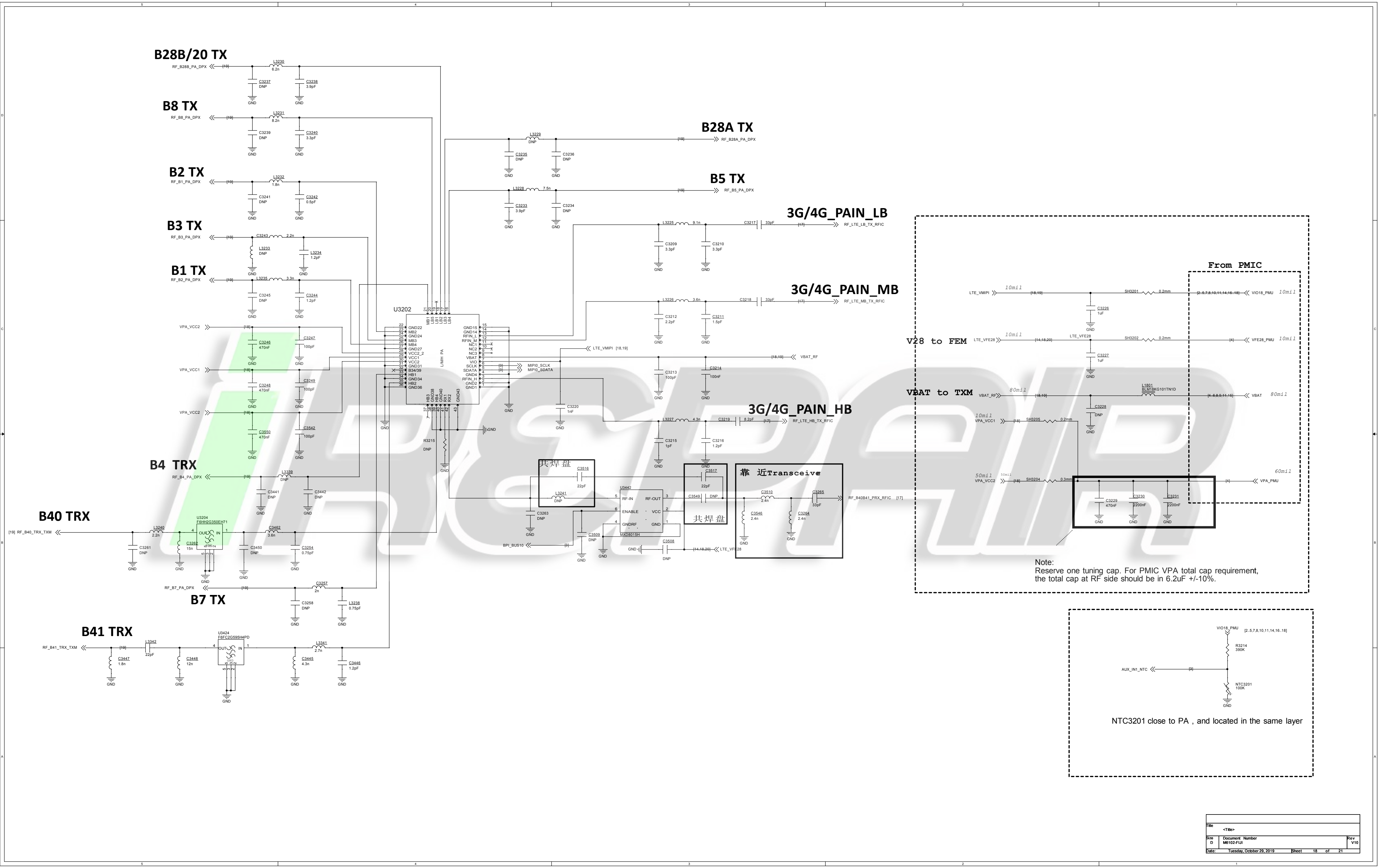

B28B/20 TX
B8 TX
B2 TX
B3 TX
B1 TX
B4 TRX
B40 TRX
B7 TX
B41 TRX
B28A TX
B5 TX
3G/4G_PAIN_LB
3G/4G_PAIN_MB
3G/4G_PAIN_HB
U3202
共焊盘
靠 近Transceive
From PMIC
V28 to FEM
VBAT to TXM
Note:
Reserve one tuning cap. For PMIC VPA total cap requirement, the total cap at RF side should be in 6.2uF +/-10%.
NTC3201 close to PA , and located in the same layer
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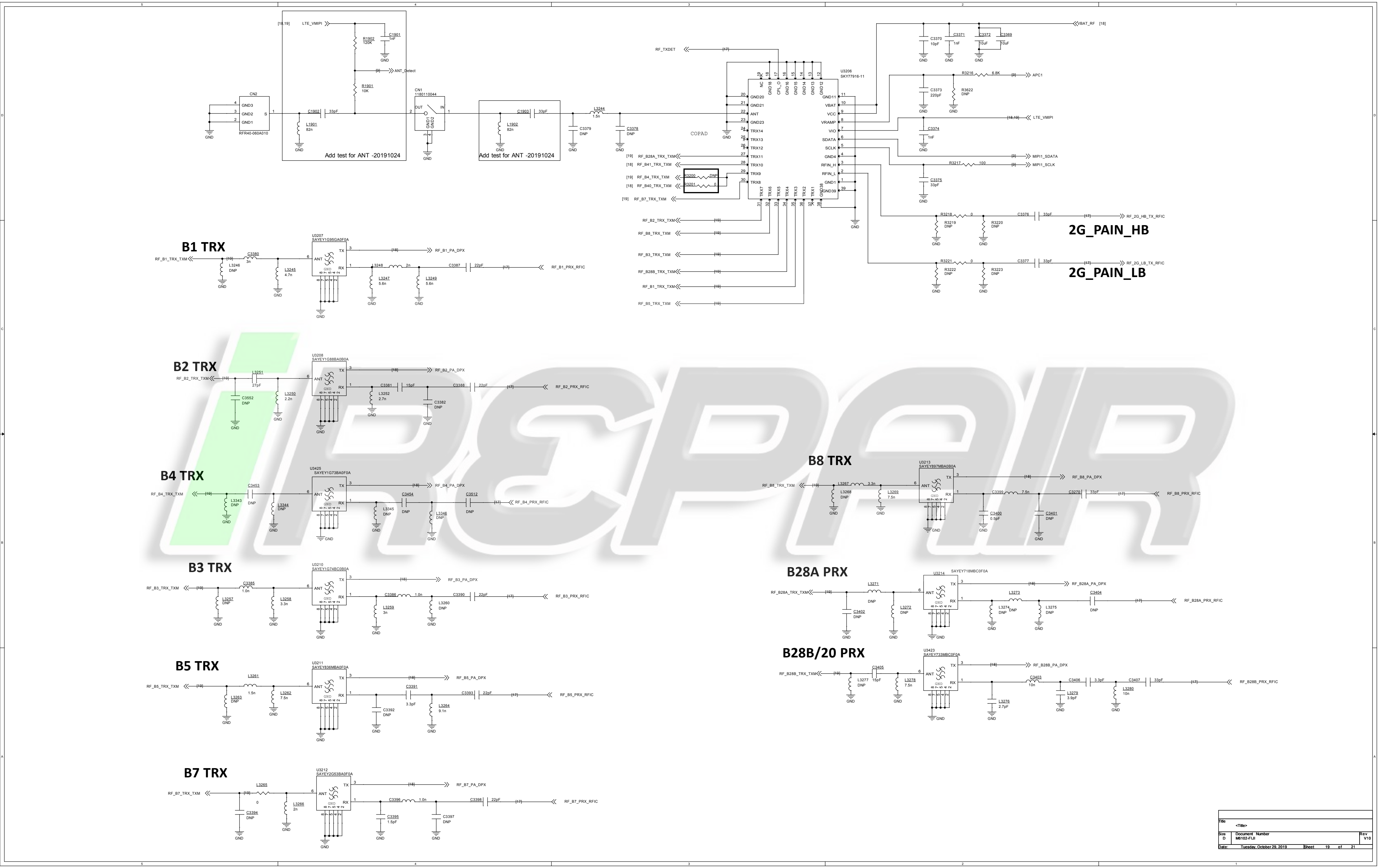

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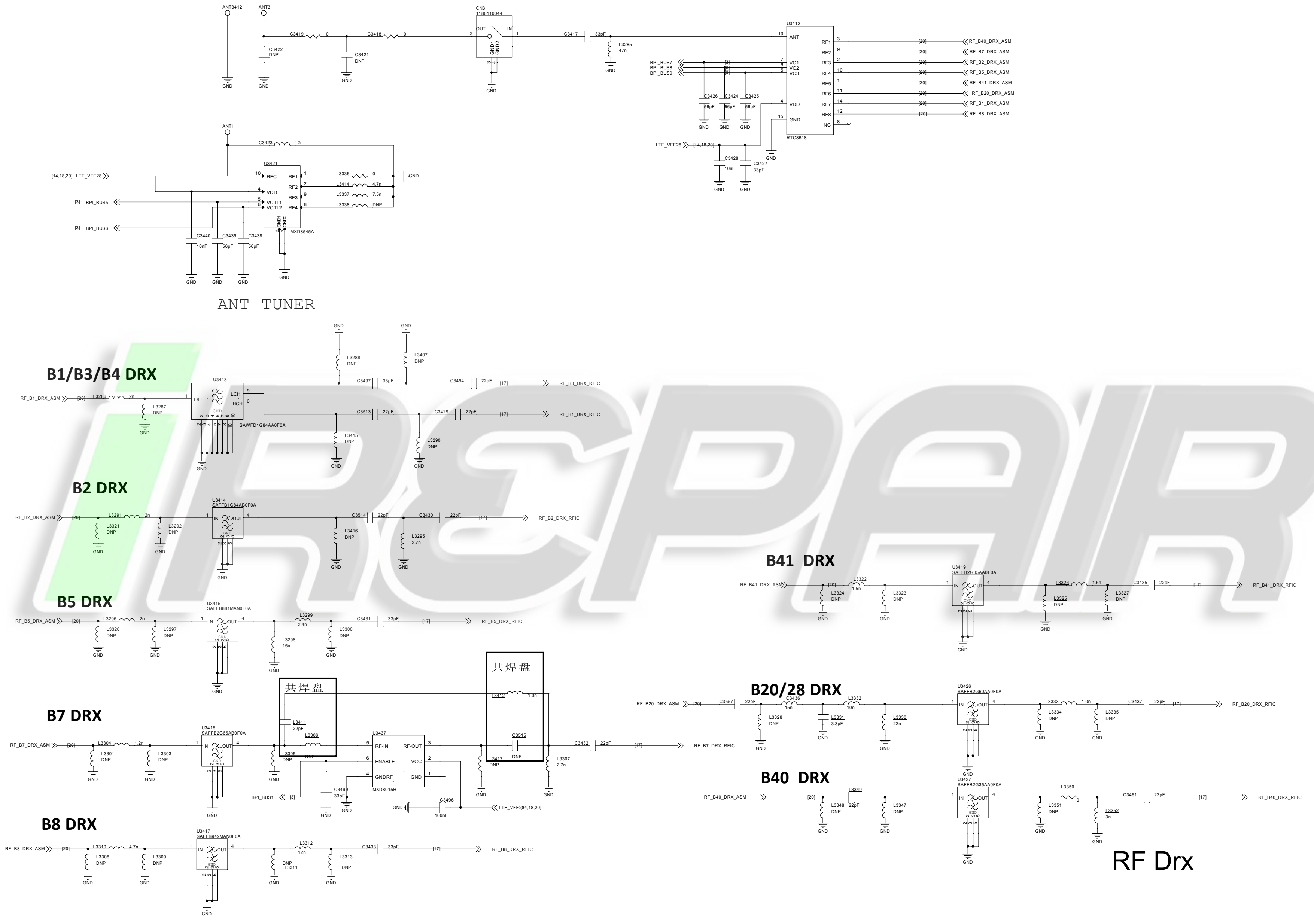

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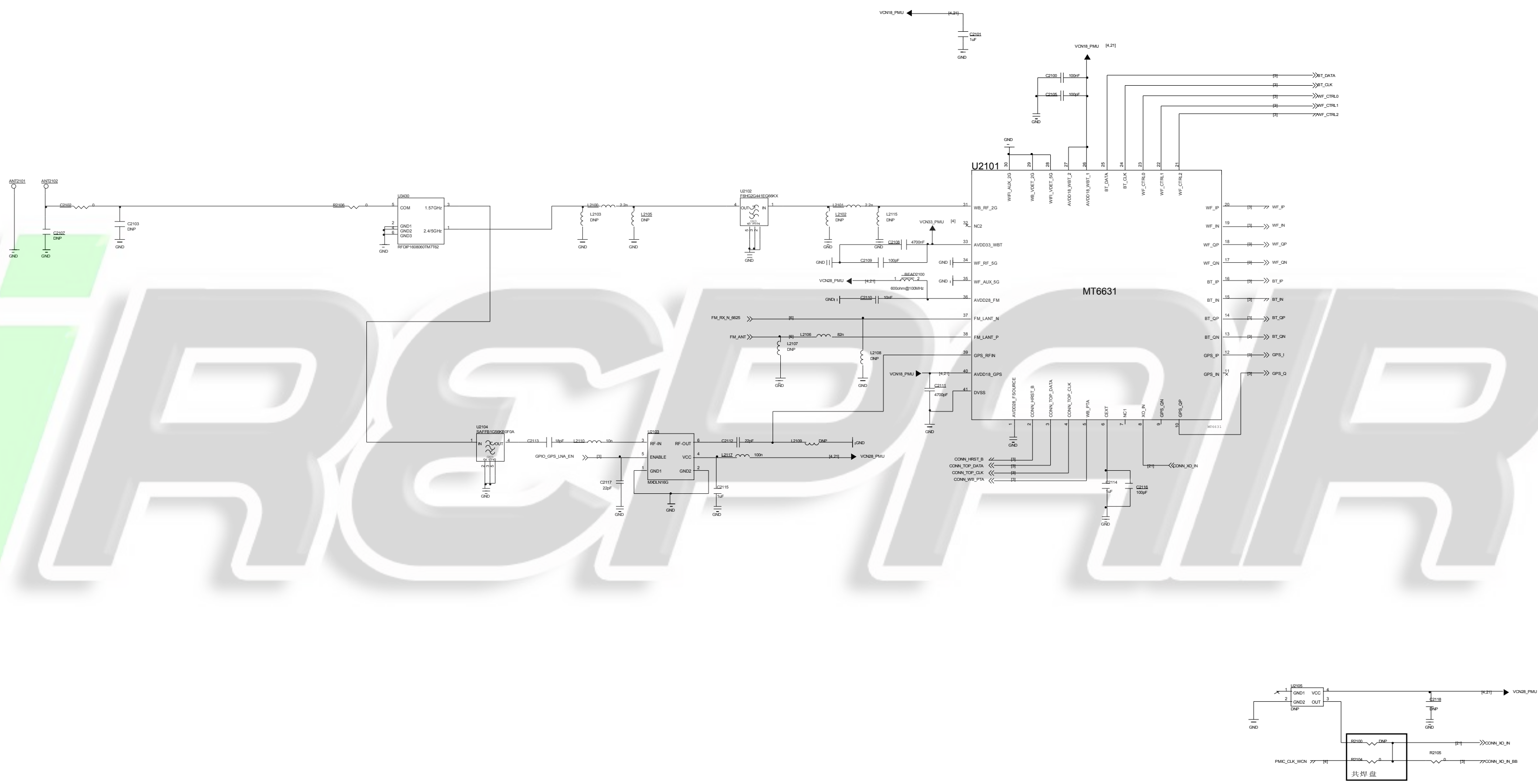

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